

Increasing prevalence of Japanese cedar pollinosis: a meta-regression analysis.

BACKGROUND: Japanese cedar pollinosis, caused by the pollen of the Japanese cedar tree (*Cryptomeria japonica*), is the commonest seasonal allergic disease in Japan. A number of epidemiological surveys have been reported on Japanese cedar pollinosis, but it has never been assessed systematically or quantitatively. To confirm the increasing prevalence of Japanese cedar pollinosis and related factors, we conducted a meta-regression analysis on population-based surveys in Japan. **METHODS:** We searched for data from population-based surveys in which serological methods were used to test all participants. Weighted regression of logit-transformed prevalence and sensitization rates were used to evaluate the effects of the year of survey, age, and degree of urbanization. We also analyzed the relationship between prevalence and sensitization rate. **RESULTS:** Thirty-eight reports with 27 subgroups for prevalence and 134 subgroups for sensitization rate were selected from the literature published in the years between 1986 and 2000. The Japanese cedar pollen sensitization rate was found to be significantly correlated with the year of survey, age, and degree of urbanization (adjusted $R^2 = 0.55$). The coefficient for the correlation between the prevalence and the sensitization rate revealed a statistically significant correlation (Pearson's $r = 0.70$, $p < 0.001$). **CONCLUSIONS:** The prevalence of Japanese cedar pollinosis among adolescents was predicted to be 28.7% in metropolitan areas and 24.5% in the general population in urban areas in the year 2004, derived from the estimated sensitization rate and the relationship between sensitization rate and prevalence. The prevalence of Japanese cedar pollinosis increased 2.6-fold between 1980 and 2000, and the prevalence differed considerably according to age and degree of urbanization. Copyright (c) 2005 S. Karger AG, Basel